

Tamir Yirga

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EDUCATION

Northeastern University

Seattle, WA

Master of Science in Computer Science | GPA: 4.0/4.0

Graduating Dec 2026

Coursework: Scalable Distributed Systems, Machine Learning, Computer Vision, Algorithms, Advanced Software Dev

Research Assistant: Co-authored an LLM hallucination study on training-data prevalence, probing layer activations

Teaching Assistant: Scalable Distributed Systems, Cloud Computing, Computer Vision – mentored grad students

EXPERIENCE

LakeSail

Jan 2026 – May 2026

Software Engineer Intern

San Francisco, CA

- Contributed 3 major PRs to a Rust-native Spark-compatible query engine, including Spark 4.1 TIME type and function support spanning the SQL parser, analyzer, Arrow conversion, and Spark Connect layers
- Owned Arrow-native execution for Python UDFs and UDTFs, removing the cross-language (Rust to Python) serialization boundary to deliver a 15x throughput improvement, and added UDF/UDTF keyword arguments
- Built a benchmarking suite comparing the Rust engine against JVM-based Spark across 5 ML/AI inference workloads and 4 execution paths, validating core use cases and identifying optimization limits

Canaria

Jul 2025 – Sep 2025

Software Engineer Intern

New York, NY

- Architected a FastAPI text-to-SQL service over a 700M+ record dataset, replacing multi-table JOINS with a pre-indexed materialized view for 50x faster PostGIS proximity and time-series queries
- Designed a 3-stage GPT-4 text-to-SQL pipeline with dynamic schema injection over a 90-day rolling window and hot-swappable prompt strategies, generating valid SQL on 99% of 225 benchmark queries
- Implemented Redis query caching to bypass expensive LLM calls, cutting repeat text-to-SQL latency from 4s to <200ms

Scandiweb

May 2022 – Aug 2024

Full Stack Engineer

Riga, Latvia

- Refactored 60% of legacy services on 8+ platforms (\$50M+ revenue) into service-oriented architectures, zero downtime
- Optimized GraphQL APIs and added Redis caching, cutting response times 30-40% for high-traffic endpoints

Endubis Labs

Apr 2021 – May 2022

Full Stack Engineer

Addis Ababa, Ethiopia

- Built a secure wallet for 7K+ users using WASM cryptography, reducing wallet sync time 60% via batched UTXO filtering

PROJECTS

InferRS (Mini LLM Inference Engine) | Rust, SIMD, GGUF, Llama | [GitHub](#)

Jan 2026 – Present

- Engineered a zero-copy Rust LLM engine with memory-mapped GGUF parsing, KV-cache, and GQA; 923 tests
- Developed INT8 quantization and parallel SIMD GEMM kernels, yielding 21x speedup, 4x memory savings

LedgerKV (Distributed KV DB) | Java, gRPC, Raft, Kubernetes, Prometheus | [GitHub](#)

Jul 2025 – Present

- Developed a distributed KV database with a custom LSM-tree storage engine and dual Raft/quorum consensus via gRPC
- Built Prometheus/Grafana monitoring and a Jepsen-style linearizability checker, cutting p99 latency via hedged requests

Distributed Ski Tracker | Java, Spring Boot, RabbitMQ, DynamoDB, Redis | [GitHub](#)

Apr 2025 – Aug 2026

- Cut DynamoDB write requests 96% (13,168 to 533) and cardinality writes 62% via batching and HyperLogLog
- Eliminated 2x replica over-admission (200 to 99.8 events/s against a 100 target) via Redis fleet coordination

TECHNICAL SKILLS

Languages & Frameworks: Rust, Java, Python, C++, Go, PHP, TypeScript, SQL, React, FastAPI

Distributed Systems: Apache Spark, Apache Arrow, gRPC, Redis, Kafka, RabbitMQ

ML/AI: PyTorch, TensorFlow, JAX, Claude Code, OpenAI API, Amazon Bedrock

Databases: PostgreSQL, DynamoDB, ClickHouse, MongoDB, Firestore, PostGIS

Cloud & Infrastructure: Kubernetes, Docker, Terraform, CI/CD, AWS, GCP, Linux, Prometheus, Grafana, Git

Certificates: [Google Cloud Certified: Professional Cloud Architect](#)

LEADERSHIP & AWARDS

Student Director, NU App Lab | Northeastern University

Fall 2025 – Present